

The Bias-Variance Tradeoff

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Introduction

Every supervised-learning error decomposes into structural bias (the wrong model class cannot fit the truth), variance (sampling noise in the fit), and irreducible noise. The bias-variance tradeoff is the single most important mental model for picking between a rigid and a flexible estimator.

Prerequisites

Regression basics; expected value and variance of estimators.

Theory

For squared loss and a target $f(x) + \varepsilon$ with $\mathbb{E}[\varepsilon] = 0$, $\text{Var}(\varepsilon) = \sigma^2$:

$$\mathbb{E}[(y - \hat{f}(x))^2] = \underbrace{(\mathbb{E}[\hat{f}(x)] - f(x))^2}_{\text{Bias}^2} + \underbrace{\mathbb{E}[(\hat{f}(x) - \mathbb{E}[\hat{f}(x)])^2]}_{\text{Variance}} + \sigma^2.$$

Complex models: low bias, high variance. Simple models: high bias, low variance. Regularisation, averaging (bagging), and dropout explicitly shift along this curve.

Assumptions

Squared loss and iid samples; trained estimator is a random function of the training set.

R Implementation

```
set.seed(2026)

true_f <- function(x) sin(x)
x_test <- seq(0, 2 * pi, length.out = 100)
f_true <- true_f(x_test)

n_reps <- 200; n_train <- 30

simulate_fit <- function(degree) {
  preds <- replicate(n_reps, {
    x_train <- runif(n_train, 0, 2 * pi)
    y_train <- true_f(x_train) + rnorm(n_train, 0, 0.3)
    mdl <- lm(y_train ~ poly(x_train, degree))
    predict(mdl, newdata = data.frame(x_train = x_test))
  })
  bias <- rowMeans(preds) - f_true
  variance <- apply(preds, 1, var)
  list(bias2 = mean(bias^2), variance = mean(variance))
}
```

```
sapply(c(1, 3, 7, 15), simulate_fit)
```

Output & Results

As polynomial degree grows, bias^2 falls and variance rises; the sum (excess MSE beyond noise) is minimised at an intermediate flexibility.

Interpretation

“Polynomial degree 3 balanced bias (0.12) and variance (0.09), minimising excess MSE; degree 15 reduced bias to 0.001 but inflated variance to 0.45, hurting generalisation.”

Practical Tips

- Overfitting corresponds to high variance (the same structure fits differently across training resamples).
- Underfitting corresponds to high bias (structurally too restrictive to capture the signal).
- Ensembling (bagging, random forests) reduces variance without increasing bias.
- Regularisation (ridge, lasso) increases bias but reduces variance – tune by CV.
- Deep learning’s “double descent” is a modern complication; the classical curve is the best starting intuition.

For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren’t.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- Cross-validation, nested CV, bootstrap .632+
- Regularisation: ridge, lasso, elastic net
- Trees, random forests, gradient boosting
- Interpretability and SHAP
- Tabular neural networks with torch
- Imaging and sequence models intro
- tidymodels pipelines
- FDR, knockoffs, replication crisis